

QUARTO PRESENTATION QUICK GUIDE

```
# Create a new slide:
## Slide Title
# Main presentation title:
# Netflix Data Analysis
# Subsection on a slide:
### Subtitle
# Insert R code:
Insert -> Code Chunk -> R
# Basic R chunk:
```

```
```{r}
code here
##render the presentation
```

**#CLEANING OF DATA**

```
Look at the data
View(netflix_clean)
str(netflix_clean)
summary(netflix_clean)
Drop a column that is not useful anymore
df$year <- NULL
Keep only rows with a condition
df_actual <- subset(df, status == "actual")
df_actual <- df[df$status == "actual",]
Replace a spelling issue / typo
df$month <- sub("Mai", "May", df$month)
df[df$month == "Mai", "month"] <- "May"
```

**#DUPLICATES**

```
Finding all the duplicates
duplicates <- duplicated(df)
sum(duplicates)
in order to keep only duplicated rows for inspection
df_duplicates <- df[duplicates,]
Remove all full duplicates
df_no_duplicates <- df[!duplicated(df),]
Find partial duplicates
dup_partial <- duplicated(df[, c("product", "region", "month")])
df_partial_duplicates <- df[dup_partial,]
Show all duplicate entries incl. first occurrence
dup_all <- duplicated(df[, c("product", "region", "month")]) |
duplicated(df[, c("product", "region", "month")], fromLast = TRUE)
df_all_duplicates <- df[dup_all,]
```

**#MISSING VALUES - NA**

```
Check where NA values are
is.na(df)
Count the NAs per column
colSums(is.na(df))
Keep rows that contain at least one NA
df_with_na <- df[rowSums(is.na(df)) > 0,]
Remove rows with NA
netflix_clean <- netflix_clean %>% filter(!is.na(release_year))
if there are missing values
replace missing values
netflix_clean$country[is.na(netflix_clean$country)] <- "Unknown"
Calculate while ignoring NA
mean(df$value, na.rm = TRUE)
```

**#MULTIPLE LINEAR REGRESSION**

```
model <- lm(
 adults ~ fawns + rainfall,
 data = df
)
summary(model)
Purpose:
Predicts one variable using multiple predictors.
Model:
$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots$
```

```
Output:
- Coefficients (Estimate)
- P-values
- R2 (Model Fit)
Interpretation:
$p < 0.05 \rightarrow$ predictor is significant
Higher R2 $\rightarrow$ model explains more variance
```

```
Example:
adults = dependent variable (Y)
fawns, rainfall = predictors (X)
```

**# CHI-SQUARE TEST**

```
table1 <- table(df$gender, df$program)
chisq.test(table1)
H0:
No association between variables
$p < 0.05 \rightarrow$ association exists.
```

**#SHAPIRO-WILK TEST (NORMALITY TEST)**

```
shapiro.test(df$variable)
Tests whether data follows a normal distribution.
H0:
Data is normally distributed.
Interpretation:
$p > 0.05 \rightarrow$ normal distribution assumed
$p < 0.05 \rightarrow$ data is NOT normally distributed
Decision:
Normal distribution $\rightarrow$ Parametric tests
(T-Test, ANOVA, Pearson Correlation)
No normal distribution $\rightarrow$ Non-parametric tests
(Wilcoxon, Mann-Whitney U, Kruskal-Wallis, Spearman Correlation)
```

**# ANOVA (ANALYSIS OF VARIANCE)**

```
result <- aov(
 empathy ~ program,
 data = data16
)
summary(result)
Purpose:
Compares means across 3 or more groups.
H0:
No difference between group means.
Interpretation:
$p < 0.05 \rightarrow$ at least one group differs
$p > 0.05 \rightarrow$ no significant difference
If ANOVA is significant:
TukeyHSD(result)
Purpose:
Shows which specific groups differ.
```

**# PREDICTIONS WITH LINEAR REGRESSION**

```
1. Create model
model <- lm(
 n ~ release_year,
 data = titles_per_year
)
2. Create new data for prediction
new_years <- data.frame(
 release_year = c(2025, 2026, 2027)
)
3. Predict values
predictions <- predict(
 model,
 newdata = new_years
)
Purpose:
Uses an existing regression model to predict future / new values.
Important:
The column name in newdata must match the model variable exactly.
Here: release_year
Interpretation:
predicted value = expected number of titles for that year
```

**#STATISTICAL TEST AND REGRESSION**

```
Normality test
shapiro.test(df$variable)
```

```
Chi-square test for two categorical variables
contingency_table <- table(df$gender, df$program)
chisq.test(contingency_table)
```

```
Correlation test
cor.test(dfx, dfy, method = "pearson") # normal data
cor.test(dfx, dfy, method = "spearman") # non-normal / ranks
cor.test(dfx, dfy, method = "kendall") # smaller samples
```

```
Linear regression
model <- lm(n ~ release_year, data = titles_per_year)
summary(model)
```

```
Visual regression line in ggplot
geom_smooth(method = "lm", se = TRUE)
```